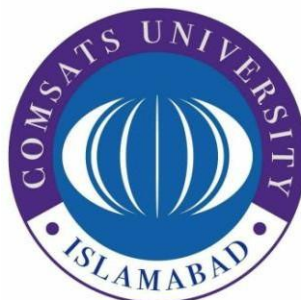


COMSATS University Islamabad

Attock Campus



Department Of Computer Science

<i>Course</i>	<i>SQE</i>
<i>Instructor</i>	<i>Mam Munnazza</i>
<i>Assignment No.</i>	<i>04</i>

Student Details

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Project Title:

Student Management System

Python Code:

```
1  students = []
2
3  def add_student():
4      roll = input("Enter Roll No: ")
5      name = input("Enter Name: ")
6      marks = input("Enter Marks: ")
7
8      student = {
9          "roll": roll,
10         "name": name,
11         "marks": marks
12     }
13
14     students.append(student)
15     print("Student Added Successfully")
16
17
18 def view_students():
19     if len(students) == 0:
20         print("No Students Found")
21
22     else:
23         for s in students:
24             print("-----")
25             print("Roll:", s["roll"])
26             print("Name:", s["name"])
27             print("Marks:", s["marks"])
28
29
30 def search_student():
31     roll = input("Enter Roll No to Search: ")
32
33     found = False
34
35     for s in students:
36         if s["roll"] == roll:
37             print("Student Found")
38             print(s)
39             found = True
40
41     if found == False:
42         print("Student Not Found")
43
44
45
46 def update_student():
47     roll = input("Enter Roll No to Update: ")
48
49     for s in students:
50         if s["roll"] == roll:
51             s["name"] = input("Enter New Name: ")
52             s["marks"] = input("Enter New Marks: ")
53             print("Student Updated")
54             return
55
56     print("Student Not Found")
57
58
59
60 def delete_student():
61     roll = input("Enter Roll No to Delete: ")
62
63     for s in students:
64         if s["roll"] == roll:
65             students.remove(s)
66             print("Student Deleted")
67             return
68
69     print("Student Not Found")
```

```
73 while True:
74
75     print("\n==== Student Management System =====")
76     print("1. Add Student")
77     print("2. View Students")
78     print("3. Search Student")
79     print("4. Update Student")
80     print("5. Delete Student")
81     print("6. Exit")
82
83     choice = input("Enter Choice: ")
84
85     if choice == "1":
86         add_student()
87
88     elif choice == "2":
89         view_students()
90
91     elif choice == "3":
92         search_student()
93
94     elif choice == "4":
95         update_student()
96
97     elif choice == "5":
98         delete_student()
99
100    elif choice == "6":
101        print("Program Closed")
102        break
103
104    else:
105        print("Invalid Choice")
```

Manual Testing :

Test Case ID:

TC-001

Steps:

1. Run the program
2. Select Add Student
3. Enter valid student details
4. Save the record

Expected Result:

Student should be added successfully.

Actual Result:

Student was added successfully.

Test Case ID:

TC-002

Steps:

1. Run the program
2. Select View Students

Expected Result:

All student records should be displayed.

Actual Result:

All student records were displayed successfully.

Test Case ID:

TC-003

Steps:

1. Run the program
2. Select Search Student
3. Enter existing roll number

Expected Result:

Student record should be found.

Actual Result:

Student record was displayed successfully.

Test Case ID:

TC-004

Steps:

1. Run the program
2. Select Search Student
3. Enter invalid roll number

Expected Result:

System should display: Student Not Found

Actual Result:

System displayed: Student Not Found

Test Case ID:

TC-005

Steps:

1. Run the program
2. Select Delete Student
3. Enter existing roll number

Expected Result:

Student record should be deleted successfully.

Actual Result:

Student record was deleted successfully.

Test Case ID:

TC-006

Steps:

1. Run the program
2. Select Add Student
3. Leave name field empty
4. Save the record

Expected Result:

System should reject empty name field.

Actual Result:

Student was added successfully even with empty name.

Test Case ID:

TC-007

Steps:

1. Run the program
2. Select Add Student
3. Enter negative marks
4. Save the record

Expected Result:

System should reject negative marks.

Actual Result:

Student was added successfully with negative marks.

Test Case ID:

TC-008

Steps:

1. Run the program
2. Add student with roll number 101
3. Add another student using same roll number

Expected Result:

System should reject duplicate roll numbers.

Actual Result:

Duplicate roll number was accepted.

Test Case ID:

TC-009

Steps:

1. Run the program
2. Select Add Student
3. Enter marks as alphabetic characters like: abc

Expected Result:

System should reject non-numeric marks.

Actual Result:

Student was added successfully with invalid marks.

Test Case ID:

TC-010

Steps:

1. Run the program
2. Enter invalid menu option

Expected Result:

System should display invalid choice message.

Actual Result:

Invalid choice message displayed successfully.

Unit testing:**Code:**

```
1  import unittest
2
3  students = []
4
5
6
7  def add_student(roll, name, marks):
8      student = {
9          "roll": roll,
10         "name": name,
11         "marks": marks
12     }
13     students.append(student)
14     return True
15
16
17  def search_student(roll):
18      for s in students:
19          if s["roll"] == roll:
20              return s
21      return None
22
23
24  def delete_student(roll):
25      for s in students:
26          if s["roll"] == roll:
27              students.remove(s)
28              return True
29      return False
30
31
32  class TestStudentSystem(unittest.TestCase):
33
34
```

```

34
35     def test_add_student(self):
36         result = add_student("101", "Ali", 90)
37         self.assertTrue(result)
38
39     def test_search_student(self):
40         add_student("102", "Ahmed", 85)
41         student = search_student("102")
42         self.assertIsNotNone(student)
43
44     def test_delete_student(self):
45         add_student("103", "Sara", 88)
46         result = delete_student("103")
47         self.assertTrue(result)
48
49     def test_search_invalid_student(self):
50         student = search_student("999")
51         self.assertIsNone(student)
52
53
54 if __name__ == '__main__':
55     unittest.main()

```

Output:

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
Python: test_system + - []
PS E:\SQE> & C:\Users\HAMMAD\AppData\Local\Programs\Python\Python314\python.exe "e:/SQE/Assignment 4 project/test_system.py"
....
-----
Ran 4 tests in 0.001s

OK
PS E:\SQE>

```

Sonar Qube :

Terminal :

```
PS E:\SQE> pysonar --sonar-host-url=http://localhost:9000 --sonar-token=sqp_626fdfa84eae3e32f1ad41262b8a4e1aa83c7c7 --sonar-project-key=Assignment-4
INFO: Sensor Python Dependency Sensor [python] (done) | time=12ms
INFO: Sensor TextAndSecretsSensor [text]
INFO: Available processors: 8
INFO: Using 8 threads for analysis.
INFO: The property "sonar.tests" is not set. To improve the analysis accuracy, we categorize a file as a test file if any of the following is true:
    * The filename starts with "test"
    * The filename contains "test." or "tests."
    * Any directory in the file path is named: "doc", "docs", "test", "tests", "mock" or "mocks"
    * Any directory in the file path has a name ending in "test" or "tests"
INFO: Start fetching files for the text and secrets analysis
INFO: Using Git CLI to retrieve dirty files
WARNING: Retrieving only language associated files, make sure to run the analysis inside a git repository to make use of inclusions specified via "sonar.text.inclusions"
INFO: Starting the text and secrets analysis
INFO: 12 source files to be analyzed for the text and secrets analysis
INFO: 12/12 source files have been analyzed for the text and secrets analysis
INFO: Sensor TextAndSecretsSensor [text] (done) | time=1002ms
INFO: ----- Run sensors on project
INFO: Sensor Zero Coverage Sensor
INFO: Sensor Zero Coverage Sensor (done) | time=49ms
INFO: ----- Gather SCA dependencies on project
INFO: Dependency analysis skipped
INFO: SCM Publisher No SCM system was detected. You can use the 'sonar.scm.provider' property to explicitly specify it.
INFO: CPD Executor 6 files had no CPD blocks
INFO: CPD Executor Calculating CPD for 6 files
INFO: CPD Executor CPD calculation finished (done) | time=33ms
INFO: Analysis report generated in 326ms, dir size=307.9 kB
INFO: Analysis report compressed in 1387ms, zip size=66.0 kB
INFO: Analysis report uploaded in 216ms
INFO: ANALYSIS SUCCESSFUL, you can find the results at: http://localhost:9000/dashboard?id=Assignment-4
INFO: Note that you will be able to access the updated dashboard once the server has processed the submitted analysis report
INFO: More about the report processing at http://localhost:9000/api/ce/task?id=d542d381-7e35-4411-bb15-a0407960578f
INFO: Analysis total time: 9.655 s
INFO: SonarScanner Engine completed successfully
PS E:\SQE>
```

Dashboard:

main

391 Lines of Code • Version not provided • [Set as homepage](#)

Quality Gate

Passed

The last analysis has warnings. [See details](#)

Last analysis 2 minutes ago

New CodeOverall Code

Vulnerabilities 0 Open issues A	Bugs 1 Open issues C	Code Smells 11 Open issues A
Accepted issues 0	Coverage 0.0%	Duplications 0.0%

Bug:

This branch duplicates the one on line 75. [python:S1862](#)

Related "if/else if" statements should not have the same condition [python:S1862](#)

Line affected: L77 • Effort: 10min • Introduced: 8 minutes ago

Open ▾ Not assigned ▾ unused ... ▸

Where is the issue? Why is this an issue? Activity

```
70
71
72
73
74
75
76
77
```

```
def fb():
    # randomness without purpose
    r=random.randint(1,10)
    if r==5:
        print("special")
    elif r==5: # duplicate condition bug
```

This branch duplicates the one on line 75.

Type: Bug Major ▾

Code Smells:

1:

Define a constant instead of duplicating this literal "Student Not Found" 3 times. [python:S1192](#)

String literals should not be duplicated [python:S1192](#)

Line affected: L42 • Effort: 6min • Introduced: 9 minutes ago

Open ▾ Not assigned ▾ design ▸

Where is the issue? Why is this an issue? How can I fix it? Activity

Assignment 4 > Assignment 4 project/Code.py [Open in IDE](#) [See all issues in this file](#)

```
37
38
39
40
41
42
```

```
print("Student Found")
print(s)
found = True

if found == False:
    print("Student Not Found")
```

Type: Code Smell Critical ▾

2:

Remove the unused function parameter "url". [python:S1172](#)

Unused function parameters should be removed [python:S1172](#)

Line affected: L25 • Effort: 5min • Introduced: 10 minutes ago

Open ▾ Not assigned ▾ unused ▸

Where is the issue? Why is this an issue? How can I fix it? Activity

```
22
23
24
25
26
27
28
29
30
```

```
print(format_string.format(person=person))

def fetch_website(urllib_version, url):
    # Import the requested version (2 or 3) of urllib
    exec(f"import urllib{urllib_version} as urllib", globals())
    # Fetch and print the requested URL
    try:
```

Uncovered code

Remove the unused function parameter "url".

Type: Code Smell Major ▾

3:

Specify an exception class to catch or reraise the exception

"SystemExit" should be re-raised [python:S5754](#)

Line affected: L32 • Effort: 5min • Introduced: 12 minutes ago

Open ▾ Not assigned ▾ error-handling ▾ +

Where is the issue? Why is this an issue? How can I fix it? Activity More info

```
30 try:
31     http = urllib.PoolManager()
    Remove the unused local variable "http".
32 except:
    Specify an exception class to catch or reraise the exception
33     print('Exception')
34
```

4:

Define a constant instead of duplicating this literal "Account not found" 5 times.

String literals should not be duplicated [python:S1192](#)

Line affected: L26 • Effort: 10min • Introduced: 12 minutes ago

Open ▾ Not assigned ▾ design +

Where is the issue? Why is this an issue? How can I fix it? Activity

```
21
22
23 def deposit(account_number, amount):
24     """Deposit money into account"""
25     if account_number not in accounts:
26         print("Account not found!")
    Define a constant instead of duplicating this literal "Account not found!" 5 times.
27     return False
```

5:

Define a constant instead of duplicating this literal "Enter Account Number: " 6 times.

String literals should not be duplicated [python:S1192](#)

Line affected: L111 • Effort: 12min • Introduced: 14 minutes ago

Open ▾ Not assigned ▾ design +

Where is the issue? Why is this an issue? How can I fix it? Activity

```
108 choice = input("Enter your choice: ")
109
110 if choice == "1":
111     acc = input("Enter Account Number: ")
    Define a constant instead of duplicating this literal "Enter Account Number: " 6 times.
112
113     name = input("Enter Name: ")
114     bal = float(input("Enter Initial Balance: "))
115     create_account(acc, name, bal)
```

6:

Specify an exception class to catch or reraise the exception

"SystemExit" should be re-raised [python:S5754](#)

Line affected: L89 • Effort: 5min • Introduced: 15 minutes ago

Open ▾ Not assigned ▾ error-handling ▾ +

Where is the issue? Why is this an issue? How can I fix it? Activity More info

```
84 # exception ignored (reliability issue)
85 try:
86     x=int(input("number:"))
87     y=int(input("number:"))
88     print(x/y)
89 except:
    Specify an exception class to catch or reraise the exception
90     pass
```

7:

Remove the unused local variable "z".

Unused local variables should be removed [python:S1481](#)

Line affected: L68 · Effort: 5min · Introduced: 17 minutes ago

Type

Code Smell Minor

Open

Not assigned

unused

Where is the issue?

Why is this an issue?

How can I fix it?

Activity

67

remove the unused local variable "x".

y=20

Remove the unused local variable "y".

68

z=30

Remove the unused local variable "z".

69

print("hello")

8:

Remove the unused local variable "y".

Unused local variables should be removed [python:S1481](#)

Line affected: L67 · Effort: 5min · Introduced: 19 minutes ago

Type

Code Smell Minor

Open

Not assigned

unused

Where is the issue?

Why is this an issue?

How can I fix it?

Activity

64

def f7():

65

unused variables (maintainability issue)

66

x=10

Remove the unused local variable "x".

67

y=20

Remove the unused local variable "y".

68

z=30

9:

Remove the unused local variable "x".

Unused local variables should be removed [python:S1481](#)

Line affected: L66 · Effort: 5min · Introduced: 20 minutes ago

Type

Code Smell Minor

Open

Not assigned

unused

Where is the issue?

Why is this an issue?

How can I fix it?

Activity

65

unused variables (maintainability issue)

66

x=10

Remove the unused local variable "x".

67

y=20

Remove the unused local variable "y".

68

z=30

10:

Remove the unused local variable "balance".

Unused local variables should be removed [python:S1481](#)

Line affected: L42 · Effort: 5min · Introduced: 21 minutes ago

Type

Code Smell Minor

Open

Not assigned

unused

Where is the issue?

Why is this an issue?

How can I fix it?

Activity

38

if account_number not in accounts:

39

print("Account not found!")

40

return False

41

balance = accounts[account_number]["balance"]

42

Remove the unused local variable "balance".

Security hotspot:

Sonar/sonarLab1.py 

Open in IDE

```
1  import os
2  import random
3
4  # Global variables (maintainability issue)
5  a = []
6  b = []
7  password = "admin123" # hardcoded password (security issue)
8
9  key = "SECRETKEY123" # hardcoded API key
10
11 def f1(x,y,z): # meaningless function name
12     if x>y:
13         return x-y/z # possible division by zero (reliability)
14     else:
15         return y-x/z
16
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Bug report:

Bug ID:

BUG-001

Bug Name:

Empty Name Field Accepted

Description:

The system allows the user to add a student without entering a name.

Steps to Reproduce:

1. Run the program
2. Select "Add Student"
3. Enter Roll Number
4. Leave Name field empty
5. Enter Marks
6. Submit the data

Expected Result:

System should display an error message like: Name cannot be empty

Actual Result:

Student record is added successfully even when the name field is empty.

Severity:

Medium

Status (Pass/Fail):

Fail

Bug ID:

BUG-002

Bug Name:

Negative Marks Accepted

Description:

The system accepts negative values for student marks.

Steps to Reproduce:

1. Run the program
2. Select "Add Student"
3. Enter student details
4. Enter marks as: -50
5. Submit the data

Expected Result:

System should reject negative marks and display validation error.

Actual Result:

Student is added with negative marks.

Severity:

Medium

Status (Pass/Fail):

Fail

Bug ID:

BUG-003

Bug Name:

Duplicate Roll Numbers Allowed

Description:

The system allows multiple students to have the same roll number.

Steps to Reproduce:

1. Add first student with roll number: 101
2. Add another student using same roll number: 101

Expected Result:

System should prevent duplicate roll numbers.

Actual Result:

Both student records are added successfully.

Severity:

High

Status (Pass/Fail):

Fail

Bug ID:

BUG-004

Bug Name:

Non-Numeric Marks Accepted

Description:

The system accepts alphabetic characters in the marks field.

Steps to Reproduce:

1. Run the program
2. Add new student
3. Enter marks as: abc

Expected Result:

System should allow only numeric values for marks.

Actual Result:

Student is added successfully with invalid marks.

Severity:

Medium

Status (Pass/Fail):

Fail

Bug ID:

BUG-005

Bug Name:

Invalid Input Handling Missing

Description:

The system does not properly handle invalid or unexpected user inputs.

Steps to Reproduce:

1. Run the system
2. Enter special characters or invalid menu choices
3. Continue program execution

Expected Result:

System should display proper validation message without issues.

Actual Result:

Program behavior becomes inconsistent due to lack of validation.

Severity:

Low

Status (Pass/Fail):

Fail

Gitpush:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell - As
PS E:\SQE\Assignment 4 project> git init
Reinitialized existing Git repository in E:/SQE/Assignment 4 project/.git/
PS E:\SQE\Assignment 4 project> git add .
PS E:\SQE\Assignment 4 project> git commit -m "Initial commit"
On branch main
nothing to commit, working tree clean
PS E:\SQE\Assignment 4 project> git config --global user.name "Your Name"
PS E:\SQE\Assignment 4 project> git config --global user.email "youremail@gmail.com"
PS E:\SQE\Assignment 4 project> git commit -m "Initial commit"
On branch main
nothing to commit, working tree clean
PS E:\SQE\Assignment 4 project> git remote add origin https://github.com/hammadkhalid-dev/Project-Student-Management-System.git
error: remote origin already exists.
PS E:\SQE\Assignment 4 project> git remote remove origin
PS E:\SQE\Assignment 4 project> git remote add origin https://github.com/hammadkhalid-dev/Project-Student-Management-System.git
PS E:\SQE\Assignment 4 project> git branch -M main
PS E:\SQE\Assignment 4 project> git push -u origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (7/7), 1.42 KiB | 726.00 KiB/s, done.
Total 7 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/hammadkhalid-dev/Project-Student-Management-System.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS E:\SQE\Assignment 4 project>
```

Github :

Project-Student-Management-System (Public)

Pin Watch 0 Fork 0 Star 0

main 1 Branch 0 Tags

Go to file Add file Code

hammadkhalid-dev Initial commit 5cd8b21 · 16 minutes ago 2 Commits

main.py	Initial commit	16 minutes ago
requirements.txt	Initial commit	25 minutes ago
test_system.py	Initial commit	25 minutes ago

README

Add a README

Help people interested in this repository understand your project.

Add a README

About

No description, website, or topics provided.

Activity 0 stars 0 watching 0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Contributors 1

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Snyk:

Security Issue 1: Open Redirect (Medium)

Priority Score: 514

Description: An Open Redirect vulnerability exists in `urllib3` version 1.26.20. This occurs when a web application accepts user-controlled input that specifies a link to an external site and uses that link in a redirect. The library fails to properly validate redirect URLs, allowing attackers to craft malicious links.

Suggested Fix: Upgrade `urllib3` to version **2.5.0** or higher. Update `requirements.txt`

The screenshot shows a Snyk vulnerability report for the package `urllib3@1.26.20`. The report indicates a priority score of 696 (MAX) and lists 5 transitive issues. The primary issue is an Open Redirect vulnerability (CWE-601) with a priority score of 514, which is a medium severity issue. The report also shows other vulnerabilities fixed by upgrading to version 2.5.0 or higher, including Allocation of Resources Without Limits or Throttling (CWE-770) and Improper Handling of Highly Compressed Data (Data Amplification) (CWE-409), both with a priority score of 659.

Issue	CWE	CVSS	Priority Score
Open Redirect	CWE-601	CVSS 6.0	514
Allocation of Resources Without Limits or Throttling	CWE-770	CVSS 8.9	659
Improper Handling of Highly Compressed Data (Data Amplification)	CWE-409	CVSS 8.9	659

Security Issue 2: Allocation of Resources Without Limits or Throttling (High)

Priority Score: 659

Description: This vulnerability exists because `urllib3@1.26.20` does not impose any limits on the amount of system resources (memory, CPU, connections) it allocates during HTTP operations.

Real-World Impact: This results in a **Denial of Service (DoS)** attack — the application becomes completely unresponsive for legitimate users. In a student management system or library system, this could take down the entire application.

Suggested Fix: Upgrade `urllib3` to version **2.6.0** or higher. Update `requirements.txt`

The screenshot shows a Snyk vulnerability report for the package `zipp@3.15.0`. The report indicates a priority score of 666 (MAX) and lists 1 transitive issue. The primary issue is an Infinite loop vulnerability (CWE-835) with a priority score of 666, which is a high severity issue. The report also shows other vulnerabilities fixed by upgrading to version 3.19.1 or higher, including Allocation of Resources Without Limits or Throttling (CWE-770) and Improper Handling of Highly Compressed Data (Data Amplification) (CWE-409), both with a priority score of 659.

Issue	CWE	CVSS	Priority Score
Infinite loop	CWE-835	CVSS 6.9	666

Security Issue 3: Improper Handling of Highly Compressed Data — Data Amplification (High)

Priority Score: 659

Description: This vulnerability, also known as a "**Zip Bomb**" or "**Decompression Bomb**" attack, occurs because `urllib3@1.26.20` does not properly handle highly compressed HTTP response data. When the library receives a compressed payload, it decompresses it without checking the final size of the decompressed data..

Suggested Fix: Upgrade `urllib3` to version **2.6.0** or higher. Update `requirements.txt`

The screenshot shows a security scanner interface for the package `requests@2.28.1`. The interface displays a "Fixable issues" tab with 4 issues, "Issues with no supported fix" with 0 issues, and "Vulnerable dependencies" with 0 issues. The "Priority Score (MAX)" is 626. Below this, there are three upgrade recommendations:

- Upgrading to `requests@2.31.0` fixes 1 issue. The issue is "Information Exposure" (CWE-200, CVSS 6.1, Priority Score 626).
- Upgrading to `requests@2.32.2` fixes 1 issue. The issue is "Always-Incorrect Control Flow Implementation" (CWE-670, CVSS 5.6, Priority Score 494).
- Upgrading to `requests@2.32.4` fixes 1 issue. The issue is "Insertion of Sensitive Information into Sent Data" (CWE-201, CVSS 5.7, Priority Score 606).

At the bottom, there is a link to "Show 1 more upgrades".

Security Issue 4: Information Exposure in Requests Library (Medium)

Affected Package: `requests@2.28.1` **CWE:** CWE-200 — Exposure of Sensitive Information to an Unauthorized Actor **CVSS Score:** 6.1 (Medium) **Priority Score:** 626

Description: The `requests` library version 2.28.1 contains an Information Exposure vulnerability. The library improperly handles HTTP redirects — when a request is redirected from an HTTPS URL to an HTTP URL, sensitive headers such as **Authorization tokens**, **session cookies**, or **API keys** are forwarded to the new (less secure) destination. This is a data leakage issue

Suggested Fix: Upgrade `requests` to version **2.31.0** or higher. Update `requirements.txt`

The screenshot shows a security scanner interface for the package `flask@2.0.0`. The interface displays a "Fixable issues" tab with 3 issues, "Issues with no supported fix" with 0 issues, and "Vulnerable dependencies" with 0 issues. The "Priority Score (MAX)" is 589. Below this, there are three upgrade recommendations:

- Upgrading to `flask@2.2.5` fixes 1 issue. The issue is "Information Exposure" (CWE-200, CVSS 7.5, Priority Score 589).
- Upgrading to `flask@2.3.2` fixes 1 issue. The issue is "Information Exposure" (CWE-200, CVSS 7.5, Priority Score 589).
- Upgrading to `flask@3.1.3` fixes 1 issue (MAJOR UPGRADE). The issue is "Use of Cache Containing Sensitive Information" (CWE-524, CVSS 2.3, Priority Score 329).

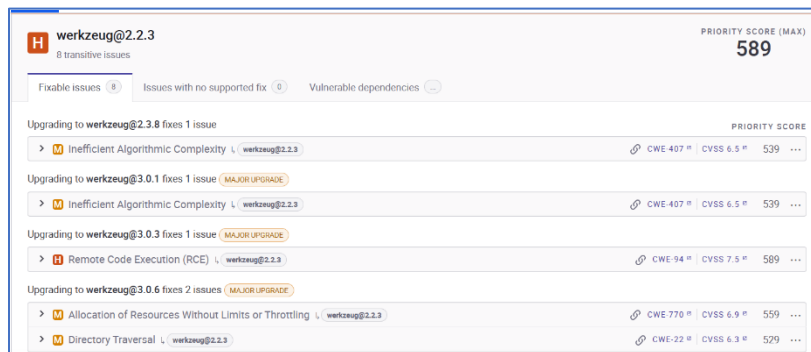
At the bottom, there is a button to "Upgrade to 2.2.5".

Security Issue 5: Remote Code Execution in Werkzeug (High)

Affected Package: werkzeug@2.2.3 **CWE:** CWE-94 — Improper Control of Generation of Code (Code Injection) **CVSS Score:** 7.5 (High) **Priority Score:** 589

Description: A **Remote Code Execution (RCE)** vulnerability exists in werkzeug@2.2.3, which is the underlying WSGI library used by Flask. This vulnerability is found in Werkzeug's **debugger feature**. When the debugger is enabled, it uses a PIN-based protection system — however, this PIN can be calculated or brute-forced by an attacker who has access to certain system information.

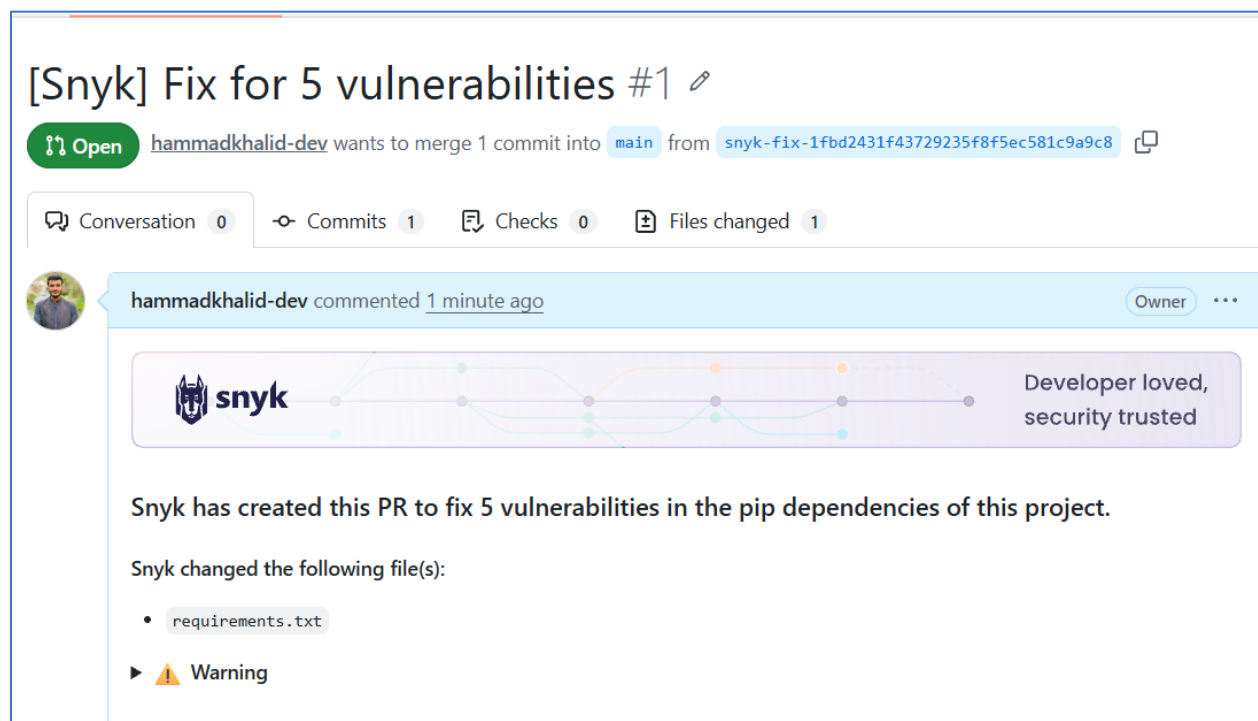
Suggested Fix: Upgrade werkzeug to version **3.0.3** or higher. Update requirements.txt



The screenshot shows a security scanner interface for the package werkzeug@2.2.3. It indicates 8 transitive issues with a priority score of 589. The interface lists several issues that can be fixed by upgrading the package:

- Upgrading to werkzeug@2.3.8 fixes 1 issue: Inefficient Algorithmic Complexity (CWE-407, CVSS 6.5, Priority Score 539).
- Upgrading to werkzeug@3.0.1 fixes 1 issue (MAJOR UPGRADE): Inefficient Algorithmic Complexity (CWE-407, CVSS 6.5, Priority Score 539).
- Upgrading to werkzeug@3.0.3 fixes 1 issue (MAJOR UPGRADE): Remote Code Execution (RCE) (CWE-94, CVSS 7.5, Priority Score 589).
- Upgrading to werkzeug@3.0.6 fixes 2 issues (MAJOR UPGRADE): Allocation of Resources Without Limits or Throttling (CWE-770, CVSS 6.9, Priority Score 559) and Directory Traversal (CWE-22, CVSS 6.3, Priority Score 529).

Fixing issues in github:



The screenshot shows a GitHub pull request titled "[Snyk] Fix for 5 vulnerabilities #1". The pull request is from the user hammadkhalid-dev and is targeting the main branch. It includes a comment from hammadkhalid-dev stating that Snyk has created this PR to fix 5 vulnerabilities in the pip dependencies of the project. The comment also mentions that Snyk changed the file requirements.txt and includes a warning icon.

[Snyk] Fix for 5 vulnerabilities #1

hammadkhalid-dev wants to merge 1 commit into main from snyk-fix-1fbd2431f43729235f8f5ec581c9a9c8

Conversation 0 Commits 1 Checks 0 Files changed 1

hammadkhalid-dev commented 1 minute ago

Developer loved, security trusted

Snyk has created this PR to fix 5 vulnerabilities in the pip dependencies of this project.

Snyk changed the following file(s):

- requirements.txt

Warning